ResQMesh · 3:30 talk · speaker script

Team Apex · Hack-Orbital 2026 · PS S.3. 6 slides, about 3:22 at 135 words a minute: 8 seconds of buffer under 3:30, then 1:30 for questions. Rehearse once with a timer; if you run over 3:30, use the "If you run long" cuts below.

CHECKPOINTS

Slide	Start	End	Words	Rubric criterion
1	0:00	0:40	89	Problem Understanding & Impact (15)
2	0:40	1:11	71	Problem Understanding & Impact (15) — what existing solutions miss
3	1:11	1:52	93	Proposed Solution Quality (25)
4	1:52	2:21	64	Innovation & Originality (20)
5	2:21	2:53	73	Technical Approach & Tech Stack (15)
6	2:53	3:22	65	Feasibility (15) · Presentation, Defence & Time Management (10)

WHAT TO SAY

SLIDE 1 0:00–0:40 · 89 words

At 2:17 on the night of 30 July 2024, a hillside in Wayanad, Kerala, let go in the dark. The only bridge was gone. The Army built a way in — in 31 hours. And survivors have hours: reported survival of trapped people falls from 88 percent on day one to 9 percent by day three. Disasters took 1.23 million lives in twenty years — one every eight minutes. In 2024, rain-driven disasters took 355 lives right here in Kerala. We can’t save the houses. Buildings can be rebuilt. People can’t.

SLIDE 2 0:40–1:11 · 71 words

So why does today’s response fail? Too far: India’s own space chief says what lies deep underground can’t be detected from space. Too slow: about fifteen hundred people searched Wayanad by hand — radar flagged signs of life, crews dug, and nobody was found. Too silent: ninety-five percent of Puerto Rico’s cell sites went dark after Hurricane Maria. Survivors isolated, rescuers operating blind. Between space and the ground, the sky is empty.

SLIDE 3 1:11–1:52 · 93 words

We fill that sky. ResQMesh: when the towers fall, the sky becomes the network. Five drones, three jobs. Two Scouts — the Eyes — fly low and find people automatically. Two Mappers — the Brain — split the zone between them, so nothing is searched twice. One Relay-Anchor — the Voice — hovers high and turns the swarm into a flying mesh. Follow one pin: a Scout sees a person and stamps a GPS location, a long-range link delivers it to the rescue base, and rescuers go to it with their own GPS. No tower. No internet. No pilot.

SLIDE 4 1:52–2:21 · 64 words

What's new? A typical search drone is flown by hand, needs towers or a direct link to its pilot, and sends back raw video that someone has to watch. ResQMesh is an autonomous swarm that re-plans when a drone is lost; the swarm itself is the network; and it delivers GPS survivor pins, not video. To our knowledge, no operational system combines all three.

SLIDE 5 2:21–2:53 · 73 words

How it works. Sense: Scouts run onboard image processing to detect people and stamp each find with GPS. Plan: Mappers run distributed area-partitioning, with energy-aware paths. Link: the drones form an ad-hoc mesh, with make-before-break handoff and store-and-forward, so pins keep flowing as batteries swap. Show: a local live dashboard, no internet. The stack is open and proven: ROS 2, Unreal Engine, a web dashboard, GPS — and drones we design ourselves in Fusion.

SLIDE 6 2:53–3:22 · 65 words

Is it feasible? Yes, because we build simulation first. Sprint one, sixteen hours: an Unreal Engine disaster, five ROS 2 drones, the planner and the dashboard. Sprint two, sixteen hours: detect, pin, relay — and survive losing a drone, live. Then custom drones and a field trial. At 2:17, nothing was in the air. Next time, something will be. Thank you — we're ready for your questions.

WHERE EACH RUBRIC CRITERION IS COVERED

Slide	Criterion	Proof on the slide
1	Problem Understanding & Impact (15)	Wayanad 02:17, 1.23 million lives, the survival cliff, Kerala's 355
2	Problem Understanding & Impact (15)	Why satellites, search on foot and cell towers all fail — in the brief's own words
3	Proposed Solution Quality (25)	ResQMesh: Eyes / Brain / Voice answer the three failures; one pin's journey
4	Innovation & Originality (20)	Autonomous swarm + swarm-as-network + GPS pins instead of video
5	Technical Approach & Tech Stack (15)	Sense / Plan / Link / Show, plus ROS 2, Unreal Engine, dashboard, GPS, Fusion
6	Feasibility (15) · Presentation, Defence & Time (10)	Two 16-hour sprints, simulation first; 6 slides, 3:22 talk, Q&A ready

Q&A CHEAT SHEET

How do the Scouts actually find people?
Onboard camera plus image processing detects people in each frame; every detection is stamped with the drone's GPS position and sent as a pin. In the demo this runs on frames from the Unreal Engine world. Honest limit: a camera sees people who are visible or partly exposed, not deep under rubble — deeper search needs extra sensors, which is the hardware phase.

How do rescuers get the coordinates if every tower is down?
The pin hops across the drone mesh to the Relay-Anchor, which holds a long-range link to a rescue base placed outside the damage zone. The base runs the dashboard locally — no internet. Rescuers navigate with their own GPS; a walkie-talkie is the fallback for the last step.

What if a drone fails or its battery dies?
Mappers re-partition the zone among the drones still flying. Planning is energy-aware, and a replacement takes over before a low-battery drone leaves (make-before-break). Pins are held with store-and-forward until a link is back. In the live demo we kill a drone on purpose.

What if the Relay-Anchor itself is lost?
The demo has one Relay, so pins are kept on the swarm and delivered when a link returns. A second Relay is the natural next step; we kept the demo to the five-drone setup.

How is this different from existing drone search and rescue?
Typical systems are piloted, depend on towers or a direct link, and return video for people to watch. ResQMesh is autonomous, the swarm is its own network, and the output is GPS survivor pins. To our knowledge no operational system combines all three.

Why simulation, and is it realistic?
In 32 hours, simulation is how we prove the whole loop — detect, pin, relay, survive a loss — with no hardware risk. The software is ROS 2, so it moves to hardware-in-the-loop later. The demo's radio mesh is a simulated link model, and we say so.

Why five drones?
It is the smallest set that shows every role with redundancy: two Scouts and two Mappers can cover for each other, plus one Relay. More Scouts and Mappers scale the search area; relays can be chained.

Where do the survival numbers come from?
The 88 / 35 / 9 percent curve is widely quoted from the 1990 Philippines earthquake; we have not traced the original paper, so the slide says "reported". The figure we verified is that 84% of survivors were rescued within the first hour (Roces et al., 1992). 1.23 million deaths is UNDRR / CRED EM-DAT.

What about safety and collisions?
The swarm flies in altitude layers — Scouts low, Mappers mid, Relay high — and Mappers assign non-overlapping zones. A real deployment would follow disaster-management and aviation rules; that is beyond Phase 1.

IF YOU RUN LONG

Skip the second sentence of slide 4 and the pin walk-through on slide 3. Never skip the closing line.

HONESTY NOTES

- 88 / 35 / 9 percent survival is labelled “reported” (1990 Philippines earthquake; original paper not traced).
- 02:17 IST is from the UN OCHA situation report; some news reports say about 1 AM (this is the small footnote on slide 1; say it if asked).
- “At 02:17, nothing was in the air” is a closing line, not a statistic: it means no autonomous search swarm was flying at that hour. Say so if challenged.
- No drone or sensor specifications are stated; hardware comes after the hackathon.
- The demo’s radio mesh is a simulated link model.